



Kids Piano

An Android app that listens to a real piano through the phone microphone and teaches a young child five white keys, one at a time.

Document	Complete app description and every screen, for review before implementation
Date	19 September 2026
Installed now	0.4.0 — one practice screen, Calibrate, Audio Lab
Designed	0.5.0 keyboard guide and export · 0.6.0 level ladder and songs · 0.7.0 falling-notes game
Screens	11 — 6 for the child, 4 for the grown-up, 1 superseded and shown for comparison
Tested on	Yamaha PSR-F52
Status	Pictures are prototypes, not screenshots of a built app. Nothing here is coded yet.

What the app is

Kids Piano turns a phone into a patient listening teacher for a child sitting at a real piano. There is no cable and no MIDI. The phone holds up the microphone, works out which note was played, and answers.

The whole app teaches **five white keys: C4, D4, E4, F4 and G4**. Nothing else appears anywhere — not in a level, not in a song, not in the game. Five keys is enough for six real tunes, and few enough that a four-year-old can own each one.

How it hears the piano

Audio arrives in blocks of 2048 samples, about 46 milliseconds each. Each block goes through pitch detection, and a note is only named once **three blocks in a row agree**. That is what stops a single noisy frame from being reported as a wrong answer, and it is why the app waits about a fifth of a second before it responds.

One physical press is one judgement. If your child holds a correct C while the app moves on to ask for D, the held C is *not* counted as a wrong D. They have to lift and press again.

The rule that shapes everything

Audio the app cannot understand is never called wrong. Two notes at once, a pitch between two keys, or a sound too quiet to read all get “I couldn’t hear that” — never a cross, never a lost star, never a miss in the game. A child must never be told they made a mistake when the microphone was the problem.

Calibration

Before a child can practise, a grown-up calibrates the piano: play each of the five keys four times while the app measures what it actually hears. It learns your piano’s real tuning and stores a profile. A PSR-F52 is close to standard pitch, but an older acoustic piano can sit noticeably sharp or flat, and without this the app would mark correct playing as wrong.

Privacy

No audio is ever recorded, stored or sent anywhere. Sound is analysed as it arrives and discarded. The only things written to disk are numbers: your piano’s measured tuning, and a log of which note was asked for and which was heard. There are no accounts, no network calls and no analytics.

The five key colours

Each key has one fixed colour, used identically on the keyboard picture, the song strip, the falling tiles and the stars — so coloured paper stickers on the real piano match the screen.

KEY	COLOUR	KEY	COLOUR	KEY	COLOUR
C4	 red	D4	 orange	E4	 yellow
F4	 green	G4	 blue	Colour is never the only signal — the note name is always shown too.	

How a session works

The ladder: eight levels, five keys

The version you installed asks for C, D, E, F and G on a child’s very first run — five new motor skills in one sitting. The ladder fixes that. A child owns one key before the second one means anything.

#	LEVEL	KEYS	PROMPTS	WHAT IT TEACHES
1	One key: C	C	6	This key is C. It is the only key lit.
2	Two keys: C and D	C D	6	C then D, in order, three times.
3	Mix it up	C D	8	The same two keys in random order — proves they read the prompt instead of repeating a pattern.
4	Three keys: E joins	C D E	8	Three in order, then five shuffled.
5	Four keys: F joins	C–F	9	Same shape.
6	All five: G joins	C–G	11	Same shape.
7	Song time	learned	song	A real tune, at the child’s own pace.
8	Falling notes	C–G	12 tiles	The game. Ships as 0.7.0.

No level can be failed. There are no lives and no timer. A level is capped at 11 prompts, because a beginner hunting for a key takes 8–20 seconds each and anything longer becomes a chore.

The shuffle is not random noise

A shuffled run is **balanced** — each key comes up about the same number of times — and it **never asks for the same key three times in a row**. Without the second rule a child can pass level 3 by hammering C and never reading the screen at all.

Stars

STARS	EARNED BY
★★★	Every prompt answered, at most one wrong key
★★	A small number of wrong keys, scaled to how long the level is
★	Getting 80% of the way through — a tired child who stops early still banks a star and still unlocks the next stop

A level unlocks when the one before it earns at least one star. Unclear audio never costs a star.

Every single key press gets an answer

WHAT YOUR CHILD PLAYED	WHAT THE SCREEN SAYS
The key that was asked for	Yes! Now D4. — or Yes! C4 again. if the next prompt is the same key
A different key	That was E4. Press C4. and the key they actually hit rings amber on screen
Right letter, wrong octave	That was C5. Press C4.
Two notes, or too quiet	I couldn’t hear that. — never counted as wrong
A press too short to identify	I heard something. Hold the key a bit longer.

The last two rows matter more than they look. Showing the wrong key beside the target is what teaches the distance between C and E. And a quick poke used to produce *nothing at all* — the previous answer just stayed on screen, which to a child looks like being ignored.

Songs at the end of a session

A song is offered as soon as every note in it is a key your child has learned. All six are traditional or original, and every note fits inside C4–G4.

SONG	KEYS NEEDED	NOTES	OFFERED AFTER
Step Up (warm-up)	C D E F	8	Level 5
Au Clair de la Lune	C D E	11	Level 4
Are You Sleeping (first half)	C D E F G	14	Level 6
Ode to Joy (first line)	C D E F G	15	Level 6
Hot Cross Buns	C D E	17	Level 4
Mary Had a Little Lamb	C D E G	26	Level 6

Two are deliberately partial: the next phrase of *Are You Sleeping* and of *Ode to Joy* both need A, which we never teach. Each stops on a complete musical phrase, so it does not sound cut off. **Songs have no rhythm requirement** — the strip waits for the child.

The eleven screens

Two halves. A child only ever sees the first six; the rest sit behind a simple maths question.

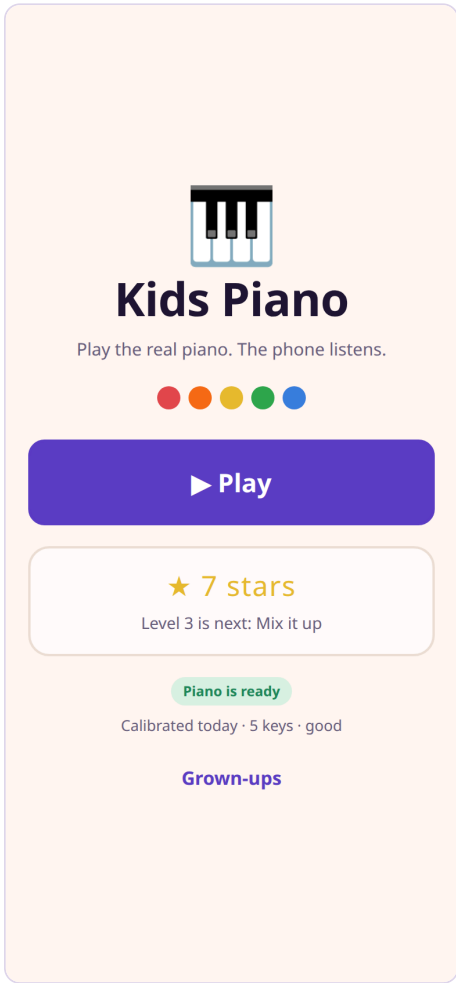
#	SCREEN	WHO	SHIPS	IN ONE LINE
1	Home	Child	0.6.0	One big Play button and the star total.
2	Level map	Child	0.6.0	Stops on a path, stars earned, locked ones dimmed.
3	Practice — 0.5.0 form	Child	0.5.0	The keyboard guide, before the ladder exists. Shown for comparison.
4	Practice — the drill	Child	0.6.0	One level of the ladder, with the wrong key ringed amber.
5	Song	Child	0.6.0	Note bubbles with lyric syllables, at the child's pace.
6	Falling notes	Child	0.7.0	Tiles drift to a line; play each as it lands.
7	Session summary	Child	0.6.0	Stars, what was learned, and the song offer.
8	Grown-ups gate	Grown-up	shipped	A maths question a young child cannot answer.
9	Calibrate	Grown-up	0.5.0	Teach the app your piano's real tuning, key by key.
10	Audio Lab	Grown-up	shipped	What the microphone is hearing, right now.
11	Files	Grown-up	0.5.0	Export the calibration and the session log.

How to read the version tags. shipped is in the 0.4.0 app you installed. 0.5.0 and 0.6.0 are designed and planned, waiting for your approval. 0.7.0 is the game, which ships after the real microphone delay has been measured from an exported log rather than estimated.

What we are deliberately not building

No cartoon mascot or character. No daily streaks or “come back tomorrow” nagging. No accounts, no cloud, no analytics. No sheet-music notation, no black keys, no chords, no two-hand playing. No tune that needs a key outside C4–G4. The app never plays the songs out loud — the piano is the instrument. And there is no fail state anywhere.

For the child · the first thing they see



Home

What it does

One enormous **Play** button, and almost nothing else. A child who cannot read should be able to start a session by pressing the only big thing on the screen.

What is on it

Play	Opens the level map at the stop they reached.
Stars	Total earned so far, and the name of the next level.
Status	Green “Piano is ready” once calibration exists; otherwise “Ask a grown-up to set up the piano first” and Play is disabled.
Grown-ups	Small text link, bottom of the screen, deliberately unglamorous.

Why it looks like this

The five coloured dots are the whole curriculum in one glance, and they match the stickers you can put on the real keys. The Grown-ups link is small on purpose: it is not for the child, and nothing behind it should look inviting to them.

For the child · chooses what to play

Let's play

Pick a stop on the path

1
One key: C
★★★

2
Two keys: C and D
★★★

3
Mix it up
★★★

4
Three keys: E joins

5
Four keys: F joins

6
All five: G joins

Song time

Falling notes · soon

Levels 1 and 2 done, 3 in progress, the rest locked

What it does

Shows the whole journey as stops on a path, so progress is visible rather than described. Tapping an unlocked stop starts that level.

What is on it

Each stop Number, name, the coloured dots for the keys it teaches, and up to three stars.

Current Outlined in purple — the one to play next.

Locked Dimmed with a padlock and not tappable. Unlocks when the stop before it earns one star.

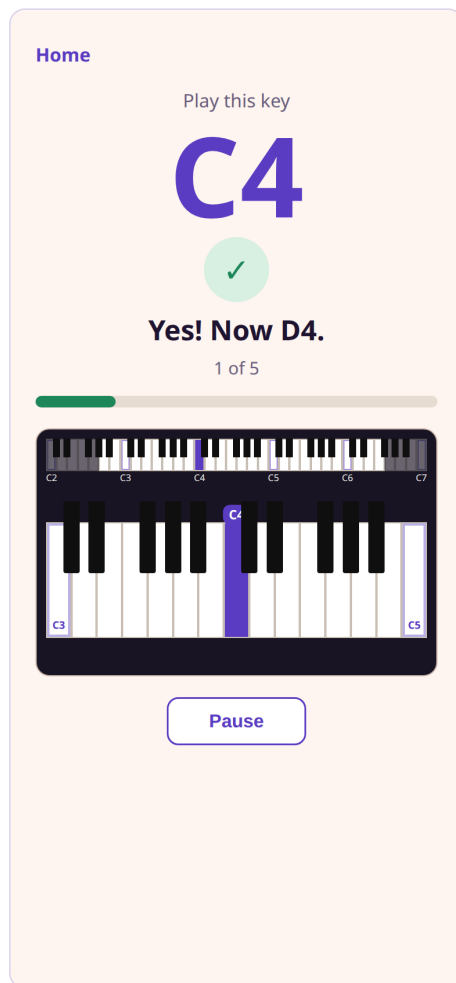
Stops 7, 8 Gold badges: a music note for Song time, a star for the game.

Notes for review

The coloured dots do real work: at a glance you can see level 4 adds yellow (E) and level 6 adds blue (G). A child learns to expect the new colour.

The eighth stop reads **"Falling notes · soon"** because the game arrives in 0.7.0. In the 0.6.0 build the ladder ends at Song time.

For the child · answers “which of the six C keys?” · superseded by screen 4



The keyboard guide, before the ladder exists

What it does

This is the direct answer to the problem you found: the app said “C” and there are six C keys on the PSR-F52. Now it says **C4**, and it shows you exactly which one.

The keyboard is two layers

Mini-map All 61 keys, C2 to C7, labelled at each C. Shows *where on the piano* to look.

Zoom strip Two octaves around the target, drawn large enough that the key name is readable and the black keys are the right shape.

A single 61-key strip was the first attempt and it failed: each white key came out about 9 dp wide, narrower than the black keys, and no label fit.

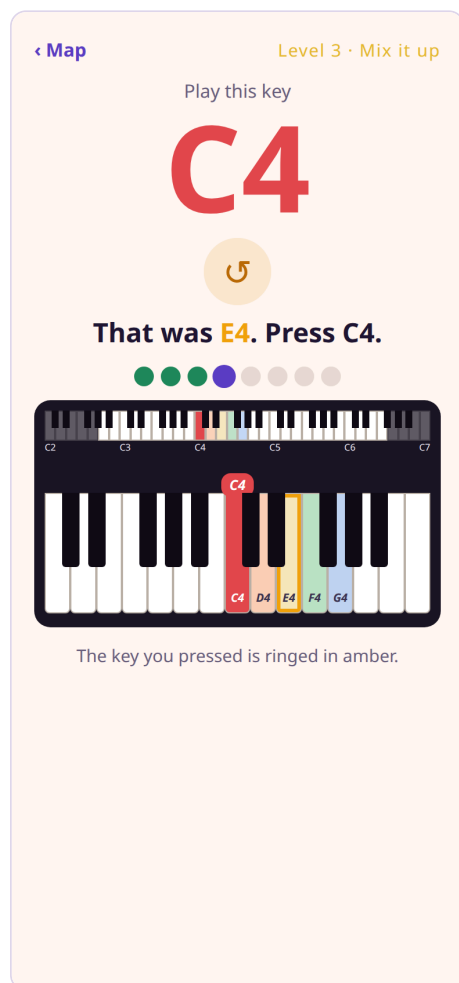
Notes for review

This screen still teaches all five keys in one run — the ladder has not arrived yet. It ships first because the export files depend on nothing else, and because it is what makes the next version testable.

4 Practice — the drill

0.6.0

For the child · levels 1 to 6 · the screen they spend most time on



C4 was asked for; E4 was played, and is ringed amber

What it does

Runs one level of the ladder. The note name fills the top in that key's colour, and the keyboard below shows both where the key is and what just happened.

What is on it

Hero	The note to play, in its own colour — red C4 here.
Badge	A tick, a retry arrow, or a soft pulse for unclear audio.
Feedback	One line naming both keys: <i>That was E4. Press C4.</i>
Pips	One dot per prompt. Green done, purple current, grey to come.
Keyboard	Target filled solid; the five taught keys tinted; the key actually pressed ringed amber.

The amber ring

This is the most important addition on the screen. Naming the wrong key in words only helps a child who reads. Lighting it up next to the target is what teaches the *distance* between C and E.

It stays lit after the finger lifts. The app internally forgets a press about a fifth of a second after release, so a ring tied to that would vanish before a child looked up.

For the child · the reward at the end of a session

MapSong · your own pace

Hot Cross Buns

Play the big note. No hurry — the song waits for you.

E4Hot

D4cross

C4buns

E4Hot

D4cross

C4buns

C4One

Yes! Now D4.

3 of 17 notes

Hot Cross Buns, three notes in

What it does

Walks through a real tune, one note at a time. This is the moment the five keys stop being an exercise and become music.

What is on it

- Bubbles** One per note, in the key's colour, with the note name inside.
- Syllables** The lyric under each bubble, so it reads as "Hot cross buns" rather than "E D C".
- Current** Enlarged and outlined; the strip scrolls to keep it centred.
- Done** Small green ticks above the notes already played.
- Keyboard** Same guide as the drill, showing where the current note lives.

No rhythm, on purpose

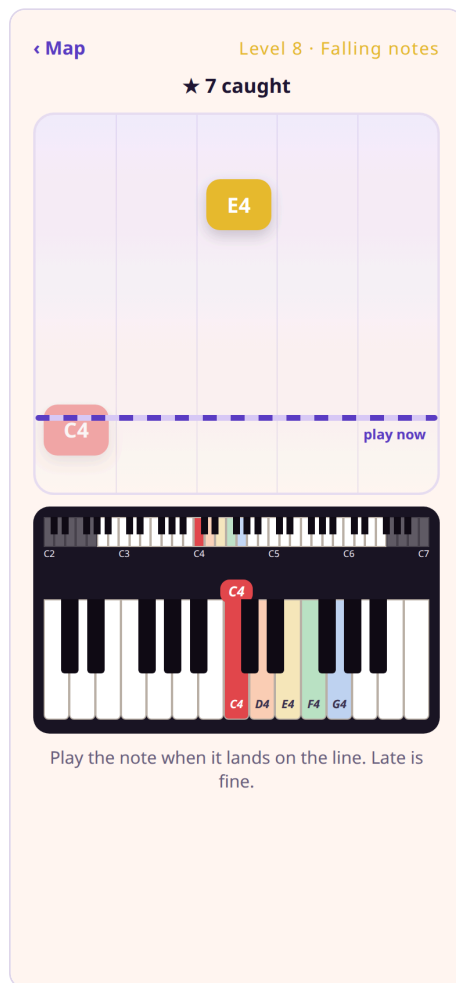
The song waits. There is no metronome and no timing score — the strip advances whenever the correct note arrives, however long that takes. A wrong note does not advance and does not end anything; it gets the same answer it would in a drill.

Rhythm is the falling game's job. Asking a five-year-old to keep time on their first tune would turn the reward back into a test.

6 Falling notes

0.7.0

For the child · level 8 · the game



One tile past the line and fading, one just spawned

What it does

Coloured tiles drift down five lanes — one lane per key, left to right in pitch order — toward a line sitting on top of the keyboard. Your child plays each key as its tile lands.

Timing, and why it is generous

Fall 3 seconds, so there is time to read the letter and find the key.

Spacing 2.5 seconds apart. Two tiles share the screen for about half a second.

Window 450 ms early, 900 ms late — deliberately lopsided.

Everything between the hammer and the app's answer costs time: the microphone and capture buffer, the first audio block that lands after the note starts, and the three agreeing blocks the detector needs. That is around 200 milliseconds and it is **always positive** — the app is never early. An even window would hand a child a second of grace before the beat and only half a second after, and beginners are late, not early.

Nothing to lose

A missed tile fades and comes back once, at the end of the run. A wrong key is named and the tile keeps falling. Unclear audio is ignored entirely. There is no health bar, no lives, no miss tally and no “game over” — the counter shows catches only.

Stars come from catches, so standing in silence scores nothing rather than a perfect run.

For the child · the end of every level

< Map

Level 4 done

★

★

★

Three keys!

C4, D4 and E4 — 11 out of 11 right.

You can play a song now

These tunes only need C, D and E — the keys you just learned.

♪ Play Hot Cross Buns

♪ Play Au Clair de la Lune

Play the level again

Back to the map

Level 4 finished clean — and two songs are now playable

What it does

Closes the session with what was earned, and immediately offers something to do with it.

What is on it

Stars	Scale in with a small burst.
One line	What was learned, in plain words: "C4, D4 and E4 — 11 out of 11 right."
Songs	Every tune that the newly learned keys make possible, most prominent first.
Then	Play the level again, or back to the map.

Why the song offer sits here

This is the payoff for the whole level. Finishing three keys is abstract; being told *you can now play Hot Cross Buns* is not. The song buttons only appear when the keys actually allow it, so the offer is never hollow.

If a level was stopped early but at least 80% answered, a star is still banked and this screen still appears. Quitting is never punished with a locked door.

For the grown-up · guards everything behind it

[← Back](#)

Grown-ups

Calibrate, Audio Lab and the export files stay here so a child cannot change the piano setup by accident.

What is 2 + 5?

ANSWER

Open

The two numbers change every time. A wrong answer just says "That is not the number." and lets you try again.

The two numbers change every time

What it does

Asks a simple addition question before opening the setup tools. It is not security — it is a speed bump that a four-year-old cannot walk over by mashing buttons.

Why it exists

Behind this gate are Calibrate, Audio Lab and the export files. A child who wandered into Calibrate and pressed Start would overwrite the tuning profile the whole app depends on, and practice would start marking correct playing as wrong.

Notes for review

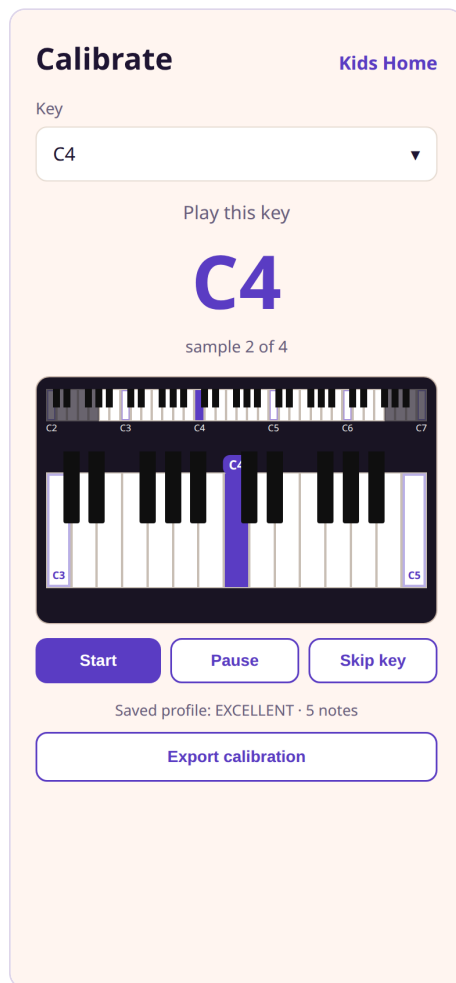
Numbers Re-rolled on every visit, so it cannot be memorised.

Wrong answer "That is not the number." and another try. No lockout, no scolding.

Keyboard Numeric only.

This screen is already in the version you installed and is unchanged by the new work.

For the grown-up · run once per piano, before a child can practise



The Key dropdown you asked for, at the top

What it does

Measures what your piano actually sounds like. Play the named key four times; the app records the pitch it hears, how much it wobbles, and how confident it is. Repeat for five keys and it has a profile.

The Key dropdown

This is the fix for the control you reported as not working — it was missing, not broken. Picking a key jumps straight to it instead of forcing you through C, D, E, F, G in order.

Range White keys from A2 to F6 — the range the pitch detector can actually hear. Offering keys outside it would let you calibrate against something the app cannot read.

Re-measuring Jumping to a key already sampled clears its samples and starts it again, so old and new readings are never mixed.

Mini-map Shows the full 61 keys with the unhearable ones faded, so the limit is visible rather than mysterious.

Also on this screen

Progress “sample 2 of 4” for the current key.

Controls Start, Pause, Skip key.

Profile Saved quality and note count — “EXCELLENT · 5 notes”.

Export Shares the calibration file directly from here.

For the grown-up · diagnostics when something seems wrong

[Home](#)Grown-ups

[Calibrate](#)[Audio Lab](#)[Files](#)

What the microphone is hearing, right now.

C4

261.9 Hz · 97% sure

Steady. This is a clean single note.

Microphone	MIC (built-in)
Sample rate	44100 Hz
Frames seen	1 284
Frames dropped	0
Signal level	0.061 RMS

[Listening](#)[Stop](#)

[Change mic](#)

Event log

CORRECT	C4	261.9 Hz	97%
AMBIGUOUS	—	two notes	41%
CORRECT	E4	329.8 Hz	94%
CORRECT	G4	392.1 Hz	96%
NO_SIGNAL	—	too quiet	—

Live note, signal metrics, and the rolling event log

What it does

Shows you exactly what the microphone is hearing, with no lesson wrapped around it. If a child insists they played the right key, this is the screen that settles it.

What is on it

Live note The note name the detector currently reads, its frequency, and how sure it is.

Hint Plain-language read on the signal: steady, too quiet, two notes at once.

Metrics Microphone in use, sample rate, frames seen, frames dropped, signal level.

Change mic Switches between the phone's audio sources — useful when one is noticeably deafer.

Event log The last judgements with status, note, frequency and confidence.

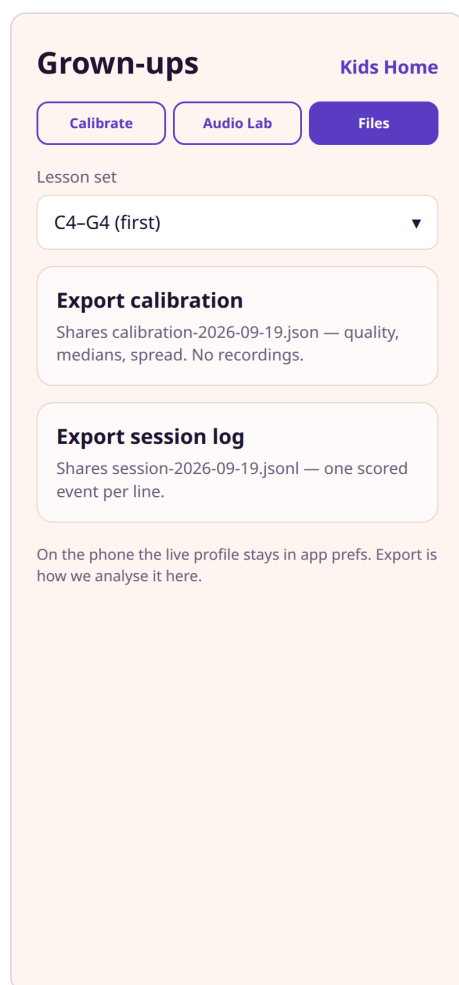
What to look for

Frames dropped above zero means the phone is struggling to keep up.

AMBIGUOUS rows usually mean the sustain pedal is down or two keys overlap. **NO_SIGNAL** means the phone is too far from the piano — on a PSR-F52 the speaker is on top, so the phone should sit near it.

This screen is already in the version you installed and is unchanged by the new work.

For the grown-up · getting the data off the phone



A third tab beside Calibrate and Audio Lab

What it does

Answers your question about where the calibration lives and how to get it out. It is a third tab in Grown-ups, not a new section of the app.

Where the data actually is

The live profile sits in the app's private preferences at `/data/data/com.vijaychhetry.kidspiano/shared_prefs/calibration.xml`. That path is unreadable on a normal phone without root, which is exactly why export exists.

The two files

Calibration `calibration-2026-09-19.json` — quality, and per key the measured pitch, the median, the spread and the sample count.

Session log `session-2026-09-19.jsonl` — one line per scored event: what was asked, what was heard, the confidence, the timestamp.

Both go out through the normal Android share sheet, so they can go to email, Drive, or anywhere else. **Neither contains any audio.**

Also here

The **Lesson set** dropdown picks which five keys practice uses — C4-G4 by default, with a lower C3-G3 set for a piano whose upper register is hard for the phone to hear.

The session log is also what sets the falling game's timing. Exporting one real run on your PSR-F52 gives the true delay between a hammer strike and the app's answer, which is the one number no test can produce.

What happens next

Three releases, in order

VERSION	WHAT LANDS	SCREENS
0.5.0	Note names with octaves, the two-layer keyboard guide, the Calibrate key dropdown, and the two export files.	3, 9, 11
0.6.0	The seven-level ladder, the six songs, key colours, stars, and an answer for every key press including ones too short to name.	1, 2, 4, 5, 7
0.7.0	The falling-notes game, with its timing set from a real measurement.	6

Why the game is last

It is the only feature whose correctness depends on how long the phone takes to hear a note, and every number in this document for that is arithmetic rather than measurement. 0.6.0 writes a session log with a timestamp on every judged press. Ship it, let your child use it next to the real PSR-F52, export the log, and the hit window can be set from your piano and your phone instead of an estimate.

How this was checked

The plan behind these screens was reviewed line by line against the existing code. That review found nine serious problems, all fixed before any code was written — among them a test that would have failed on its first run, a scoring rule that gave three stars to a child who played nothing at all, and a hit window that put its generosity on the wrong side of the beat.

Implementation is test-first: 48 new automated checks, each one required to fail if the feature it guards is broken. The rules that matter most — unclear audio is never wrong, a held key cannot answer twice, no level can be failed — each have their own test.

What I need from you

Look at the eleven screens and tell me what is wrong with them. Specifically worth your attention: is the drill screen (4) readable enough for your child at arm's length; does the song screen (5) make sense without sheet music; is the level map (2) obvious enough that they can choose for themselves; and is the star rule generous enough that finishing feels good.

Nothing is built yet. Once you approve, 0.5.0 and 0.6.0 go in that order, each with the full test suite run and an installable APK, plus a short README on exporting the two files.